

Experiment- 02

AIM:-

1. List all distinct job in Employee.
2. List all information about employee in Department Number 30.
3. Find all department number with department names greater than 20.
4. Find all information about all the managers as well as the clerks in department 30.
5. List the Employee name, Employee numbers and department of all clerks.
6. Find all managers not in department 30.
7. List information about all Employees in department 10 who are not manager or clerks.
8. Find Employees and jobs earning between 1200 and 1400.
9. List Name and Department Number of employee who are clerks, analyst or salesman.
10. List Name and Department Number of employee whose names began with M.

PROCEDURE:-

- SELECT – retrieve data
- DISTINCT – remove duplicate values
- WHERE – filter records
- AND / OR – combine conditions
- IN / NOT IN – check multiple values
- BETWEEN – range condition
- LIKE – pattern matching

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Perform Following Query using Employee Table:-

1. List all distinct job in Employee.

Ans-

```
SELECT DISTINCT job FROM Employee;
```

```
MariaDB [2cse25_1650]> SELECT DISTINCT job
-> FROM Employee;

+-----+
| job   |
+-----+
| CLERK |
| SALESMAN |
| MANAGER |
| ANALYST |
| PRESIDENT |
+-----+

5 rows in set (0.001 sec)
```

2. List all information about employee in Department Number 30.

Ans-

```
SELECT * FROM Employee WHERE deptno = 30;
```

```
MariaDB [2cse25_1650]> SELECT *
-> FROM Employee
-> WHERE deptno = 30;

+-----+-----+-----+-----+-----+-----+-----+-----+
| EMPNO | ENAME | JOB       | MGR  | HIREDATE | SAL      | COMM     | DEPTNO |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 7499  | ALLEN | SALESMAN | 7698 | 1981-02-20 | 1600.00 | 300.00   | 30     |
| 7521  | WARD  | SALESMAN | 7698 | 1981-02-22 | 1250.00 | 300.00   | 30     |
| 7654  | MARTIN | SALESMAN | 7698 | 1981-09-28 | 1250.00 | 1400.00  | 30     |
| 7698  | BLAKE | MANAGER | 7839 | 1981-05-01 | 3135.00 | NULL     | 30     |
| 7844  | TURNER | SALESMAN | 7698 | 1981-09-08 | 1500.00 | 0.00     | 30     |
| 7900  | JAMES | CLERK    | 7698 | 1981-12-03 | 1045.00 | NULL     | 30     |
+-----+-----+-----+-----+-----+-----+-----+-----+

6 rows in set (0.003 sec)
```

3. Find all department number with department names greater than 20.

Ans-

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SELECT deptno, dname FROM Department WHERE deptno > 20;

```
MariaDB [2cse25_1650]> SELECT deptno, dname
-> FROM Department
-> WHERE deptno > 20;
+-----+-----+
| deptno | dname      |
+-----+-----+
|      30 | SALES      |
|      40 | OPERATIONS |
+-----+-----+
2 rows in set (0.001 sec)
```

4. Find all information about all the managers as well as the clerks in department 30.

Ans-

SELECT * FROM Employee WHERE deptno = 30 AND job IN ('MANAGER','CLERK');

```
MariaDB [2cse25_1650]> SELECT *
-> FROM Employee
-> WHERE deptno = 30
-> AND job IN ('MANAGER','CLERK');
+-----+-----+-----+-----+-----+-----+-----+-----+
| EMPNO | ENAME | JOB      | MGR  | HIREDATE | SAL      | COMM | DEPTNO |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 7698  | BLAKE | MANAGER  | 7839 | 1981-05-01 | 3135.00 | NULL | 30     |
| 7900  | JAMES | CLERK    | 7698 | 1981-12-03 | 1045.00 | NULL | 30     |
+-----+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.012 sec)
```

5. List the Employee name, Employee numbers and department of all clerks.

Ans-

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SELECT ename, empno, deptno FROM Employee WHERE job = 'CLERK';

```
MariaDB [2cse25_1650]> SELECT ename, empno, deptno
-> FROM Employee
-> WHERE job = 'CLERK';
```

ename	empno	deptno
SMITH	7369	20
ADAMS	7876	20
JAMES	7900	30
MILLER	7934	10

4 rows in set (0.001 sec)

6. Find all managers not in department 30.

Ans-

SELECT * FROM Employee WHERE job = 'MANAGER' AND deptno <> 30;

```
MariaDB [2cse25_1650]> SELECT *
-> FROM Employee
-> WHERE job = 'MANAGER'
-> AND deptno <> 30;
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7566	JONES	MANAGER	7839	1981-04-02	3272.50	NULL	20
7782	CLARK	MANAGER	7839	1981-06-09	2695.00	NULL	20

2 rows in set (0.012 sec)

7. List information about all Employees in department 10 who are not manager or clerks.

Ans-

SELECT * FROM Employee WHERE deptno = 10 AND job NOT IN ('MANAGER','CLERK');

```
MariaDB [2cse25_1650]> SELECT *
-> FROM Employee
-> WHERE deptno = 10
-> AND job NOT IN ('MANAGER','CLERK');
```

Empty set (0.001 sec)

8. Find Employees and jobs earning between 1200 and 1400.

Ans-

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SELECT ename, job FROM Employee WHERE sal BETWEEN 1200 AND 1400;

```
MariaDB [2cse25_1650]> SELECT ename, job
-> FROM Employee
-> WHERE sal BETWEEN 1200 AND 1400;
+-----+-----+
| ename  | job    |
+-----+-----+
| WARD   | SALESMAN |
| MARTIN | SALESMAN |
| ADAMS  | CLERK    |
+-----+-----+
3 rows in set (0.012 sec)
```

9. List Name and Department Number of employee who are clerks, analyst or salesman.

Ans-

SELECT ename, deptno FROM Employee WHERE job IN ('CLERK','ANALYST','SALESMAN');

```
MariaDB [2cse25_1650]> SELECT ename, deptno
-> FROM Employee
-> WHERE job IN ('CLERK','ANALYST','SALESMAN');
+-----+-----+
| ename  | deptno |
+-----+-----+
| SMITH  | 20     |
| ALLEN  | 30     |
| WARD   | 30     |
| MARTIN | 30     |
| SCOTT  | 40     |
| TURNER | 30     |
| ADAMS  | 20     |
| JAMES  | 30     |
| FORD   | 20     |
| MILLER | 10     |
+-----+-----+
10 rows in set (0.001 sec)
```

10. List Name and Department Number of employee whose names began with M.

Ans-

SELECT ename, deptno FROM Employee WHERE ename LIKE 'M%';

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```
MariaDB [2cse25_1650]> SELECT ename, deptno  
-> FROM Employee  
-> WHERE ename LIKE 'M%';
```

ename	deptno
MARTIN	30
MILLER	10

```
2 rows in set (0.001 sec)
```

```
MariaDB [2cse25_1650]> |
```